Detailed Analysis of Sensors and Actuators

Robotics Engineering Fundamentals | Systems Integration

Technical Presentation

Vision Sensors

Environmental Perception

A vision sensor is a device that collects visual information from its surroundings and processes it to perform complex activities.

- **Object Detection:** Identifying entities in space.
- **Recognition:** Classifying objects based on features.
- **Tracking:** Following moving targets.

Includes Camera sensors for image capture and Infrared (IR) sensors for object detection and tracking.



Tactile and Temperature Sensors

Tactile Sensors

Collects data from physical interactions with the environment. Designed after the biological sense of cutaneous touch.

- **Force Sensors:** Measures load or force applied.
- **Touch Sensors:** Mainly used in touch screens and robotic pads.

Temperature Sensors

Detects changes in temperature through a resistor or thermocouple.

- Used in industrial applications.
- Maintains set temperature limits in robotic internal enclosures.

Range and Proximity: Ultrasonic

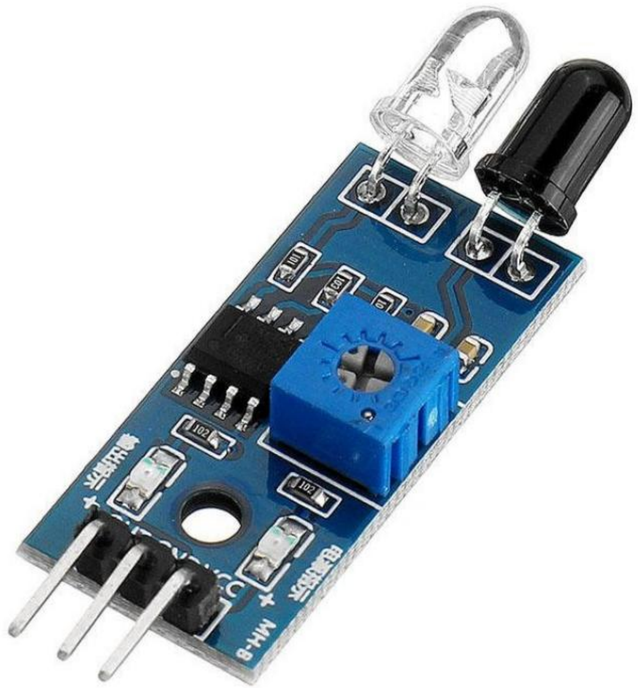
Ultrasonic Sensors

These sensors use sound waves to detect objects and measure the distance to a target object.

- Non-contact distance measurement.
- Commonly used in parking sensors.
- Provides distance by measuring time-of-flight of sound waves.



Range and Proximity: IR Sensors



Proximity Sensors

Detects the presence of an object when it enters the sensor's field using light (IR), sound, or magnetic fields.

- Used in mobile phones to detect ear proximity.
- Essential for home appliances and obstacle avoidance.
- Works without physical contact.

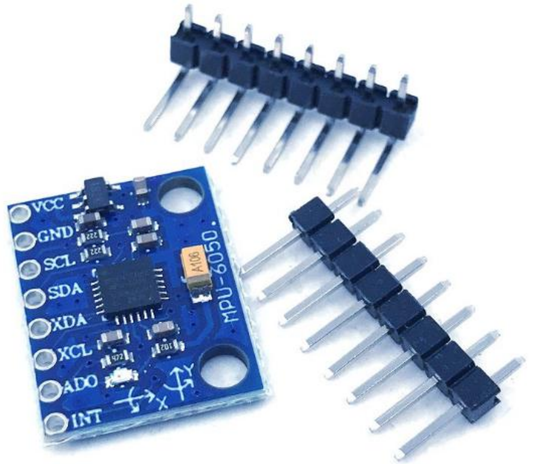
Motion Sensors: Accelerometer & Gyro

Accelerometer

Measures acceleration and tilt. Essential for auto-rotation in smartphones and track steps in wearables.

Gyroscope

Measures angular velocity. Vital for drones, virtual reality systems, and stability in gaming controllers.



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Navigation Sensors



Global Positioning System (GPS)

Provides information on position, velocity, and direction for travel by ocean, road, or air.

- Used in autonomous vehicles and drones.
- Connects to a network of satellites for geolocation.
- Critical for long-range robot navigation.

Speech and Audio Processing

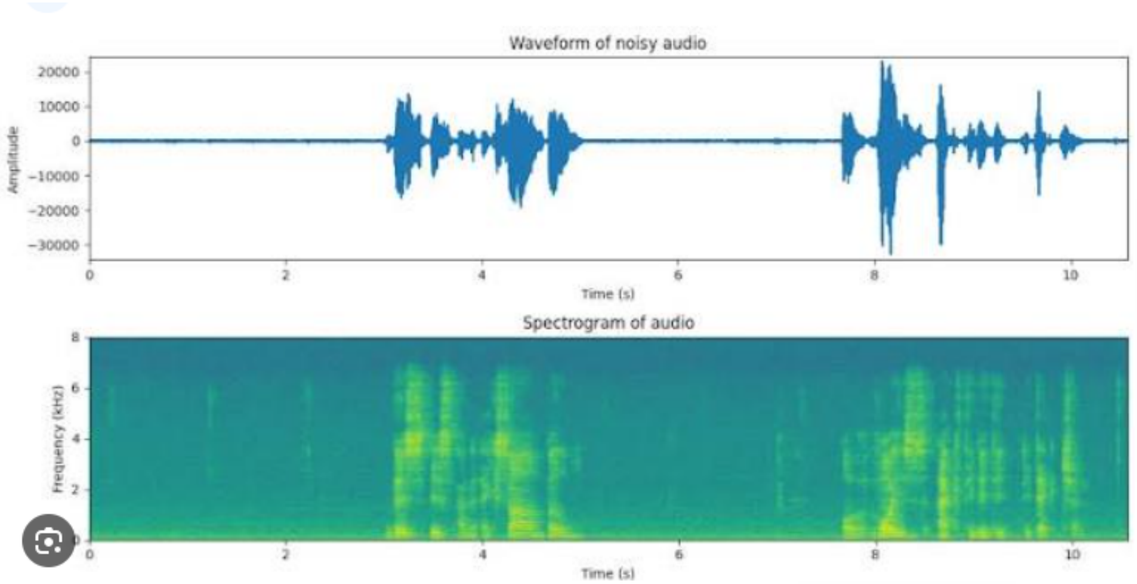
Microphone Array

Collects audio signals for voice commands and communication. Essential for voice-controlled robots.



Voice Activity Detector

Detects the presence of speech in audio signals. Helps in filtering background noise for voice assistants.



Velocity and Acceleration Sensing

Velocity Sensors

Measures the rate of change of location over time. Monitors object speed or rotation velocity.

- **Tachometers:** Monitors rotation of wheels.
- **Doppler Radar:** Measures speed via frequency shifts.





Core Functions of Sensors

Transform

Converts physical parameters like temperature or pressure into electrical impulses.

Quantify

Precisely measures the value of the physical parameter perceived.

Detect

Identifies changes or occurrences in the environment that can be utilized.

Monitor

Constantly reports on changes in the measured parameter over time.

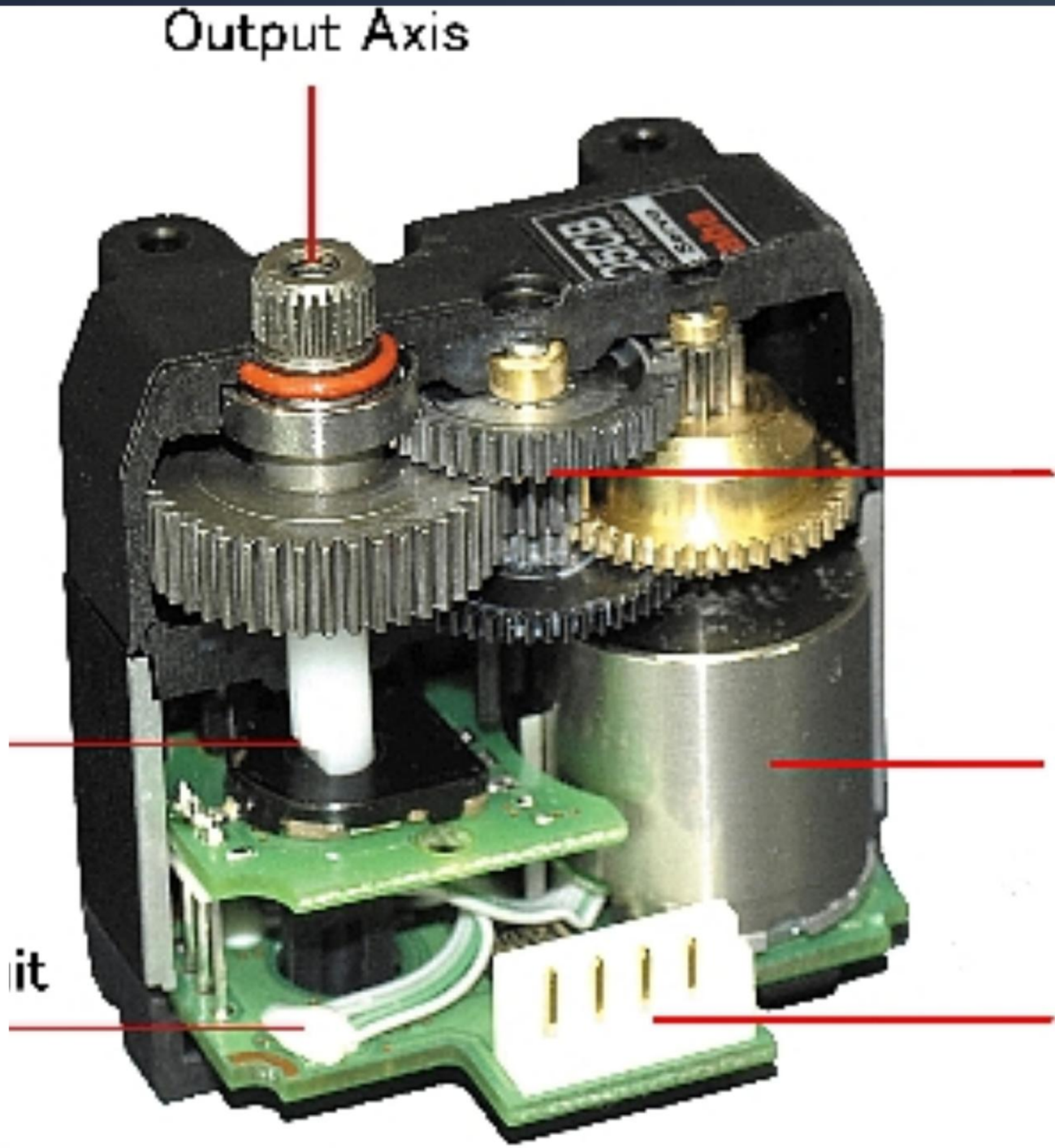
Feedback

Provides data to control systems to maintain settings or adjust to changes.

Actuators: From Signal to Motion

Actuators are devices that convert electrical, hydraulic, or pneumatic impulses into **physical motion or force**.

> *They are the necessary components that play an important role in achieving movement and functionality in robotics and aerospace.*



Questions?

Thank you for attending this detailed overview of robot sensing.

Source: Components of Robots as a System | Robotics Unit 4

Image Sources



https://image.made-in-china.com/202f0j00OEhBJmFnGHrI/Hikrobot-Mv-Sc3013xc-12m-Wbn-1-3MP-Industrial-Machine-Vision-Sensor-with-60fps-Global-Shutter-and-Profinet-Camera.webp

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https://www.sparkfun.com/media/catalog/product/cache/f8158826193ba5faa8b862a9bd1eb9e9/1/5/15569-Ultrasonic_Distance_Sensor_-_HC-SR04-01a.jpg

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